

This guide outlines how to compile multiple `.ino` files into a single `.uf2` binary using the **Arduino IDE**. This process is common for modern microcontrollers like the **Raspberry Pi Pico (RP2040)** or **ESP32-S2/S3**.

Phase 1: Preparation & Organization

The Arduino IDE treats multiple files as a single project "sketch".

1. **Folder Setup:** Place all six `.ino` files in a single folder. The folder name **must match** the name of the main `.ino` file.
 2. **Tab Order:** Open the main `.ino` file. The other files will automatically appear as tabs at the top.
 - **Logic:** The IDE concatenates (merges) these tabs into one large file behind the scenes.
 - **Sequence:** It starts with the main file and then appends the others in **alphabetical order**.
 - **Tip:** To control the merge order, rename tabs with a numeric prefix (e.g., `10_Global.ino`, `20_Setup.ino`).
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Phase 2: Compiling to Binary

Standard "Uploading" via the arrow button often bypasses the creation of a permanent file you can save. To get a file you can drag-and-drop, use the **Export** feature.

1. **Select Board:** Go to **Tools > Board** and select your specific device.
2. **Verify Code:** Click the **Checkmark** (Verify) icon to ensure no errors exist in any tab.
3. **Export:** Select **Sketch > Export Compiled Binary**.
 - This creates a subfolder in your project directory (usually called `build`).
 - Depending on your board, this folder will contain `.bin`, `.elf`, or `.hex` files.

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NOTES: after exporting it creates the build folder which has some subfolders and includes the UF2 file that is uploaded to the device. If you want to make a modification you need to dump everything into a new base folder or rename the existing build folder so it doesn't get overwritten

Phase 3: Converting to UF2

A `.uf2` file is specifically designed for "USB Flashing." Not all board packages generate this automatically.

Option A: Automatic Generation (Easiest)

Many modern board packages (like the **Raspberry Pi Pico Mbed** or **Earle Philhower** cores) automatically create a `.uf2` file in the build folder alongside the other binaries when you click "Export Compiled Binary".

Option B: Using a Converter Tool

If the IDE only gives you a `.bin` or `.hex` file, you need a utility to convert it.

- **Utility Needed:** `uf2conv.py` (a Python script provided by Microsoft/Adafruit).

- **Step-by-Step:**

1. Install **Python 3** on your computer.
2. Open a Terminal or Command Prompt in the folder containing your binary.
3. Run the command:

```
python uf2conv.py firmware.bin -c -f [FamilyID]
```

(Replace `firmware.bin` with your file name. The "FamilyID" for RP2040 is `0xe48bff57`).

Phase 4: Flashing the Device

1. **Enter Bootloader Mode:** Plug in your device while holding its **BOOT** or **RESET** button.
2. **Mount Drive:** A new drive will appear on your computer (e.g., `RPI-RP2` or `CPLAYBOOT`).
3. **Deploy:** Drag your `.uf2` file and drop it onto that drive.
4. **Auto-Run:** The drive will disappear, and the device will reboot automatically with your new code.

Summary Checklist

- **Software:** Arduino IDE.
- **Utility:** Python 3 (for manual conversion).
- **Key Command:** **Sketch > Export Compiled Binary**.
- **File Location:** Found via **Sketch > Show Sketch Folder**.